

Joseph Zhang

Software Engineer

josephzh0731@gmail.com | +1 (312) 313-9206 | Chicago, IL 60607 | [LinkedIn](#) | [Github](#)

SUMMARY

Full Stack Engineer with 8+ years of experience in **JavaScript** and **Python** engineering. Worked on building user-friendly and scalable software solutions and participated in several enterprise-level apps for various industries. Designed and implemented highly efficient microservices architectures, cloud-based solutions, and analytics-driven platforms - a big fan of TypeScript & CRM specialist.

EDUCATION

Illinois Institute of Technology 2015 – 2016
Master's Degree in Computer Science

Northwestern University 2010 – 2014
Bachelor's Degree in Economics

SKILLS

Languages: JavaScript/TypeScript, HTML5/CSS, Node.js, PHP, Go, Java, Python, C#, JSON, C++
Frameworks: Angular, React JS (Redux, Next.js), Vue.js (Nuxt.js), Ember.js, Express.js, Nest.js, Django, Flask, ASP.NET, GraphQL, D3.js
Databases: SQL, NoSQL, MySQL, PostgreSQL, MongoDB, Amazon DynamoDB
Tools: Docker, Jira, Webpack, Babel, NPM, Yarn, PNPM, GitHub, GitLab, Trello, Figma, VS Code, Jest, Mocha, Enzyme, Cypress, Playwright
Cloud Platforms: AWS, GCP, Azure
Other Skills: Accessibility, Web Socket, Google Analytics (GA), Google Tag Manager (GTM), Agile Methodology, Scrum, DevOps, CI/CD, Unit Testing, Responsive Design & Mobile-First Websites, WordPress, Headless CMS, Shopify, Storybook, SEO, Debugging, Object-Oriented Programming (OOP), Project Management, Multitasking, Teamwork, Detail-Oriented, Creative Thinking

EMPLOYMENT HISTORY

Conversant, LLC 04/2019 – Present
Software Engineer | Full Stack Developer

- Conducted in-depth research on customer engagement strategies and CRM trends using Python, resulting in the development and presentation of the report *Understanding Rental Demand: CRM Insights from Market Trends Through 2021, Illinois* to Conversant's investment team.
- Architected a scalable microservice-based project using Django and PostgreSQL, integrated with an Angular admin portal, and deployed on AWS (ECS, RDS, S3, API Gateway), improving deployment efficiency by 40%.
- Developed and integrated a CRM system into the admin portal using React.js, implementing business-critical features and building a complex D3.js-powered dashboard for investment managers, increasing user engagement by 30%.
- Built key booking and advertising modules in the user portal of Open Oasis using Next.js (React.js), TypeScript, leading to a 20% increase in platform traffic over three months.

- Collaborated with the data science team to design, develop, and maintain scalable ETL pipelines using Apache Spark and Python, processing over 10 million records daily with 99.9% uptime.
- Designed and deployed a Django-based web UI with Python to help internal clients track user profile generation in Conversant's data pipeline, reducing manual tracking effort by 60%.
- Developed responsive and interactive user interfaces using React.js, leveraging its component-based architecture and state management solutions like Redux to create scalable and maintainable web applications.
- Analyzed transaction data using Python to uncover revenue-driving factors—bookings, transactions, and search patterns—leading to data-backed strategy changes that boosted revenue by 15%.
- Presented a session on data-driven CRM strategy at Fal.Con 2023 in Charlotte, NC, attended by over 100 professionals.
- Championed test-driven development across multiple projects and mentored junior developers.

Skills: React.js, Redux, Next.js, Angular, Python, Django, Tailwind CSS, Node.js, Docker, Kubernetes, AWS...

Conversant, LLC

10/2017 – 04/2019

Associated Software Engineer

- Worked on building an Angular 7 front end with a Node.js back end and implementing Docker containerization to streamline deployments, reducing setup time by 40%.
- Participated in the development of Urbanmix Tech's MVP using React.js, GraphQL, Flask (Python), and Firebase, creating an intuitive hotel management platform that improved client demo conversion rates by 22%.
- Engineered a full lifecycle metrics reporting system using the ELK stack and Kibana, enabling real-time analytics and reducing debugging time by 35%.
- Developed Hadoop mappers for a high-throughput data ingestion pipeline handling millions of ad market transactions daily, and contributed to the initial setup of an Apache Spark and Kafka pipeline for scalable ad data processing.
- Worked directly with the PeopleSoft Team to ensure a seamless transition from the MAS90 (Sage 100 Standard) legacy accounting system, reducing data reconciliation errors by 50%.
- Modernized a legacy publisher advertising spend React UI and overhauled the Docker-based deployment process, increasing build reliability by 30%.
- Designed and developed a proof-of-concept AWS Bedrock solution for an AI-powered product support bot, aiming to reduce triage time for production issues by over 40%.
- Created a time management tool using React.js for Conversant's internal event calendar, improving event coordination efficiency by 25%.
- Collaborated with the data science team to solve high-impact challenges, demonstrating strong analytical and problem-solving skills in fast-paced development cycles.

Skills: Java, Spring Boot, Kafka, React.js, TypeScript, Node.js, MongoDB, GCP, Jest, Cypress...

Google

09/2016 – 10/2017

Software Engineer Intern

- Developed a RESTful API and implemented a backend framework to collect and analyze data from an internal test automation system, enabling efficient reporting of regression test results and improving QA turnaround time by 35%.
- Took full responsibility for maintaining and enhancing Bolt, an internal Laravel-based application, managing development tasks, resolving bugs, and optimizing performance using Laravel, JavaScript, and SQL.
- Implemented shared C/C++ common service libraries used across multiple internal tools, standardizing functionality and reducing redundant development across product teams by 25%.
- Collaborated closely with senior engineers to integrate reporting capabilities into the continuous testing pipeline, contributing to the overall stability and reliability of the software delivery process.
- Participated in code reviews and sprint planning sessions, gaining hands-on experience with agile development practices and real-world software engineering workflows.

Skills: C/C++, PHP, Laravel, jQuery, MySQL, HTML, CSS, ES6...

PROJECTS

VENUE WEST CHICAGO

<https://www.venuewestchicago.com/>

Venue West Chicago is a management platform for venue rentals, offering functionality for both venue providers and customers. integrations for CRM and analytics features to support better business decisions.

- Conducted research on CRM strategies and trends, presentation of the report ***Understanding Rental Demand: CRM Insights from Market Trends Through 2021, Illinois*** to Conversant's investment team.
- Led development of CRM module using Django/Python, Angular, and PostgreSQL, implemented core features like customer verification, lead management, campaign organization, discount management, etc.
- Implemented scraping and data collecting application for decision making for investors, identified main factors of revenue growth, resulting in 4% growth of customers in the very first 3 months after launching.

URBANMIX

<https://www.urbanmix.tech/>

URBANMIX is a real estate software platform designed to bring the reuse of existing buildings into the smart city era. It features a unique interactive 3D model system powered by a machine learning engine to automate decision-making and maximize returns for developers.

- Led the development of the MVP for a hotel management platform using React.js (Material UI + Tailwind CSS) and Flask, integrating 3D visualizations with Three.js.
- Built and launched the beta version, introducing new features for managing hotel rooms based on real-time rental status.
-